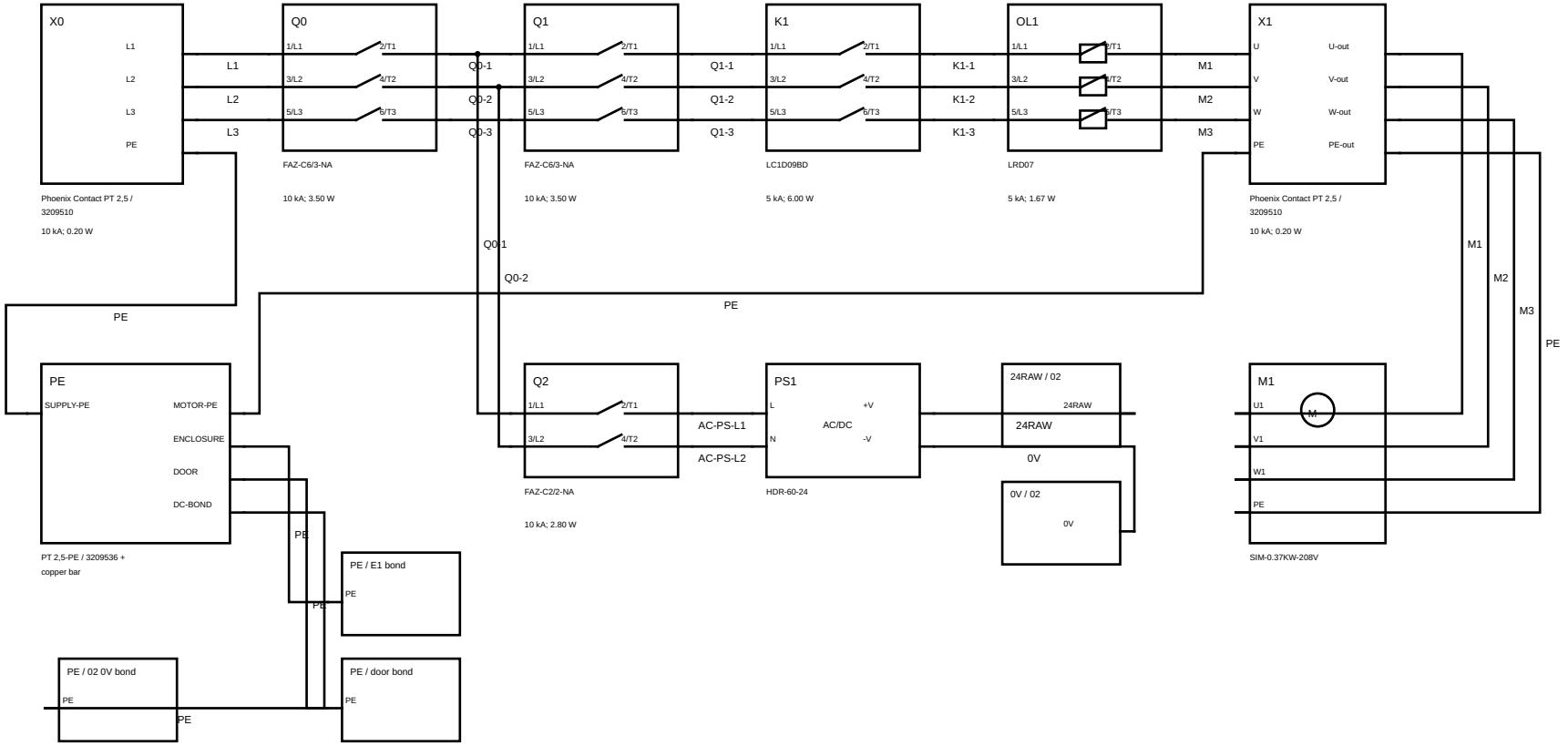
BOTTLE FILLER / 01 / Incoming power and motor starter

Revision B | Jack Lai | Simulation / engineering study | 2026-11-05



208 V, 3 phase, 60 Hz, grounded-wye only. Assumed available fault: 4 kA rms symmetrical.

Q0: incoming breaker. Q1: motor short-circuit protection. No selective trip coordination claimed.

OL1: 1.6-2.5 A range; simulated M1 nameplate 2.0 A; manual reset. Motor loss is outside enclosure.

Power: 14 AWG Cu design choice. PE: green/yellow, never switched or fused. No neutral required.

Concept study only. Upstream lockout/isolation, guarding and safety-rated stop need engineered review.

Reference conductors by net name across sheets. Crossings have no junction unless they share a terminal.

Author: Jack Lai	Incoming power and motor starter	File: bottle-filler
Date:		Folio: 1/5

BOTTLE FILLER / 02 / 24 V DC distribution and bonding

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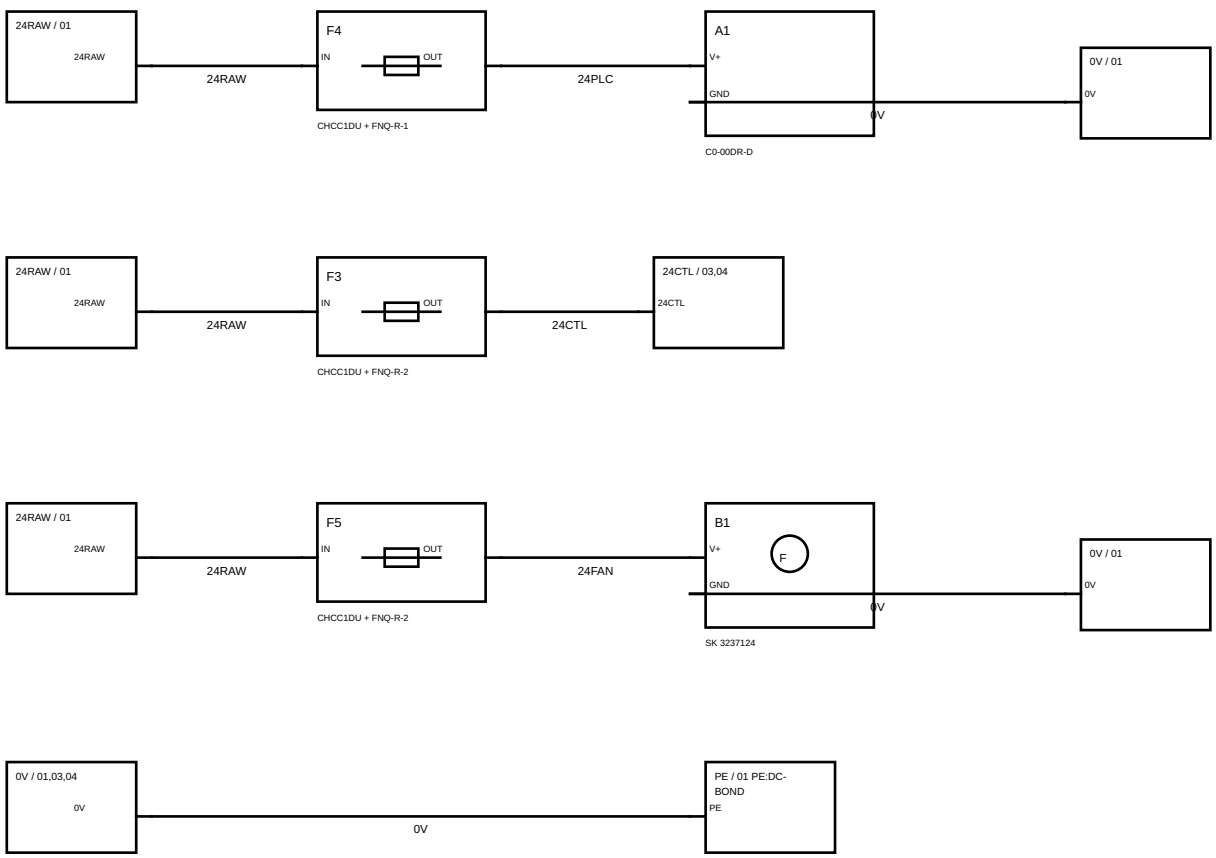

PS1 is fed line-to-line at 208 V (01). 24 V / 2.5 A supply; all field loads included in DC budget.

F4: 1 A PLC; F3: 2 A control/field; F5: 2 A fan. Control conductors: 18 AWG Cu design choice.

0 V has one intentional bond to PE at the DC distribution terminal; protective earth stays continuous.

PLC V+ and GND power logic only. Relay output commons are separately wired on 04.

B1 is continuously supplied. EF1/EF2 outlets give 16 m3/h installed, not 20 free-air.

Reference conductors by net name across sheets. Crossings have no junction unless they share a terminal.

Author: Jack Lai	24 V DC distribution and bonding	File: bottle-filler
Date:		Folio: 2/5

BOTTLE FILLER / 03 / PLC inputs and field terminal strip

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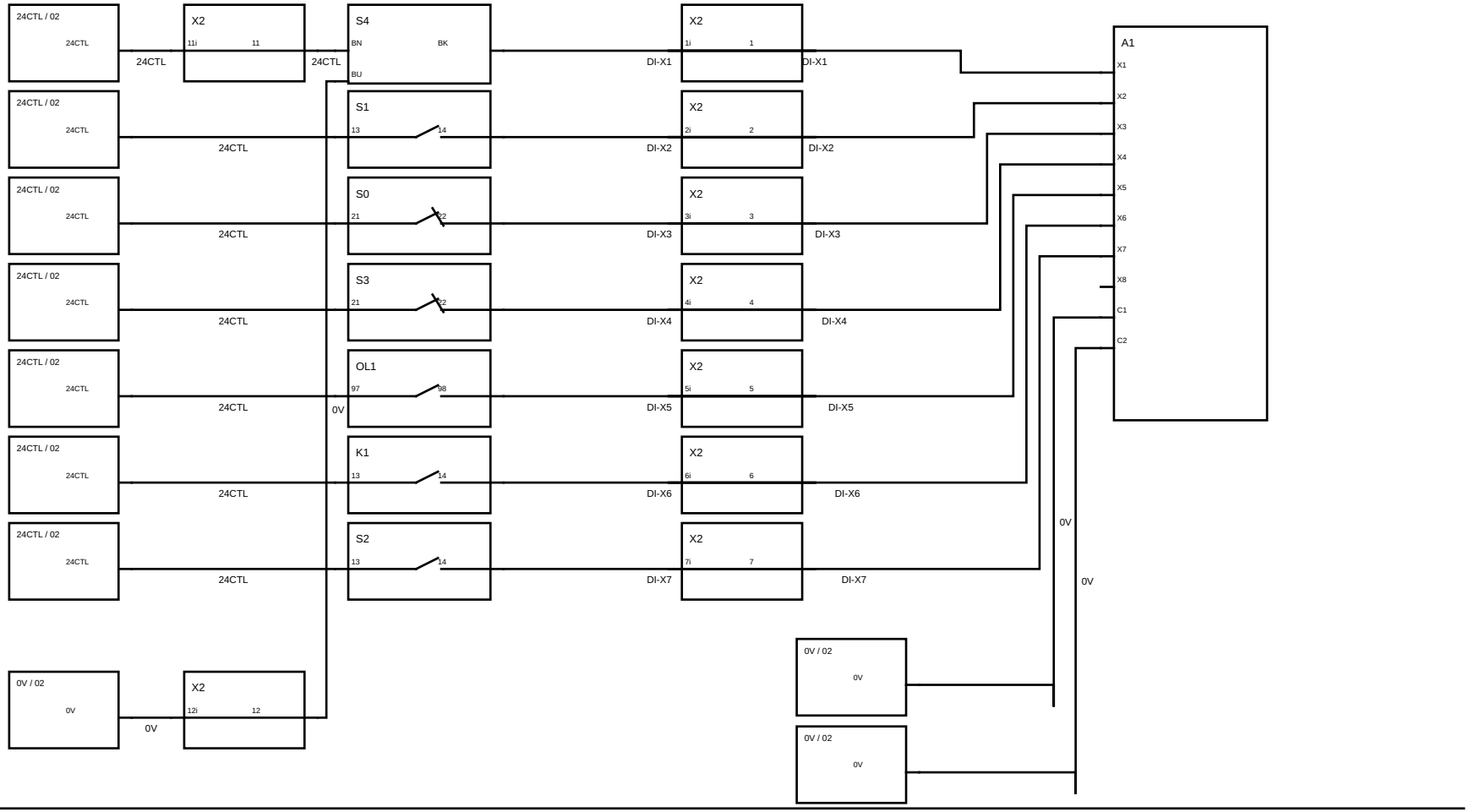

A1: C0-00DR-D. X1 bottle present; X2 start; X3 stop healthy; X4 enable healthy;

X5 overload tripped; X6 contactor feedback; X7 reset; X8 spare (unconnected). C1/C2 to 0 V.

S0 and S3 use separate NC monitoring poles here. Their hardwired interrupt poles are on 04.

S4 brown=24CTL, blue=0V, black=PNP signal. X2 terminal strip: numbered internal and field sides.

Stop and enable are educational interlocks only, not an emergency-stop or safety function.

Reference conductors by net name across sheets. Crossings have no junction unless they share a terminal.

Author: Jack Lai	PLC inputs and field terminal strip	File: bottle-filler
Date:		Folio: 3/5

BOTTLE FILLER / 04 / Hardwired inhibit and PLC relay outputs

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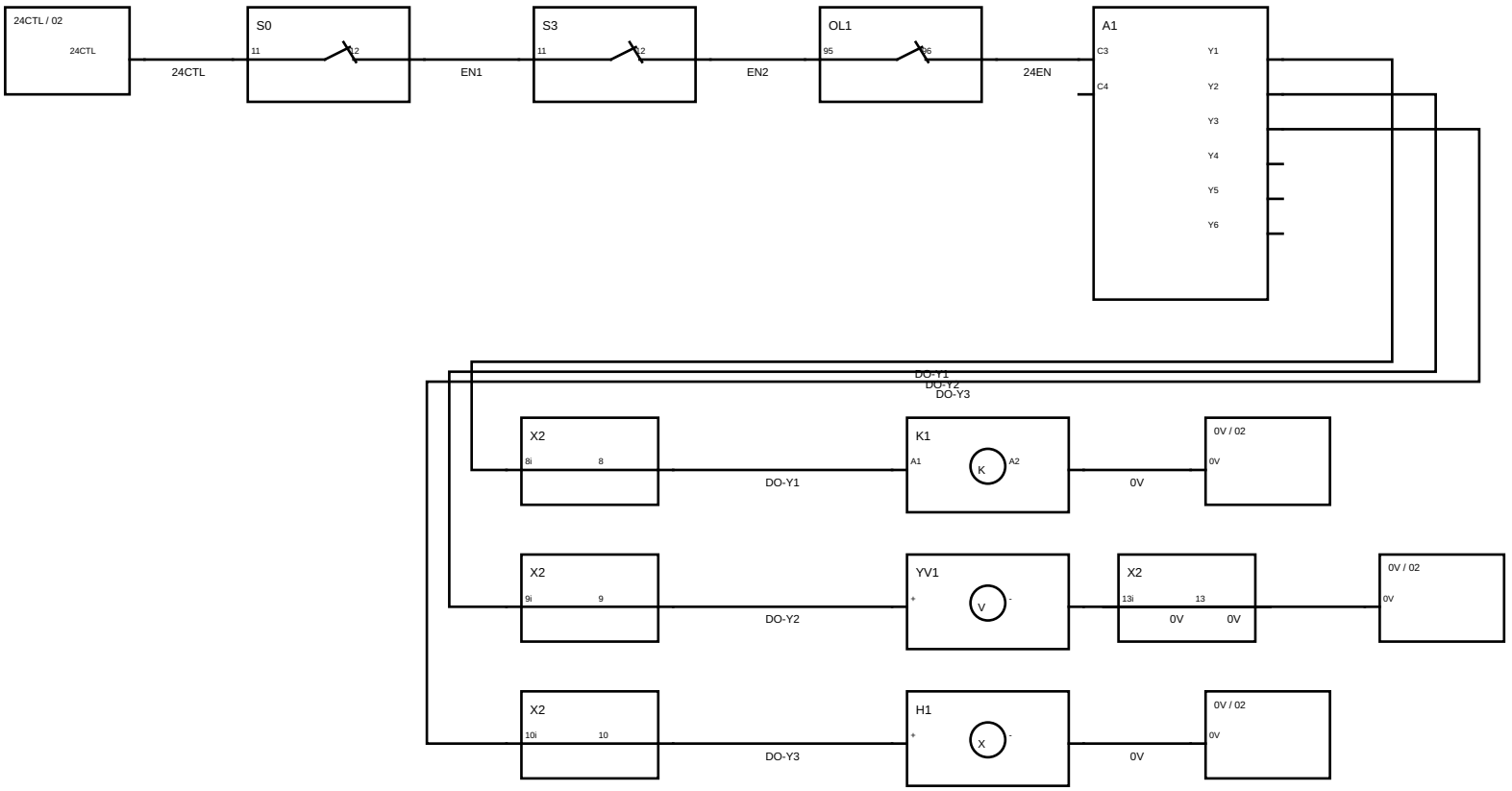

24CTL -> S0 NC -> S3 NC -> OL1 95-96 NC -> A1 C3. Opening any contact removes Y1-Y4 load supply.

A1 relay group C3 supplies Y1-Y4; C4 (Y5/Y6) unused. Y4/Y5/Y6 are spare and unconnected.

Y1 energizes K1 coil (A1/A2), Y2 opens YV1, Y3 lights H1. All loads return to 0 V.

K1 BD has built-in suppression. YV1 diode cathode at +24. X2:11/12 sensor power; X2:13 valve return.

Start latches a cycle; bottle stops conveyor and starts a 2 s fill. Stop/enable/overload aborts; reset then start.

No automatic restart. See python3 -m panel simulate for executable timing and fault sequence.

Reference conductors by net name across sheets. Crossings have no junction unless they share a terminal.

Author: Jack Lai	Hardwired inhibit and PLC relay outputs	File: bottle-filler
Date:		Folio: 4/5

BOTTLE FILLER / 05 / Panel layout, dimensions in mm

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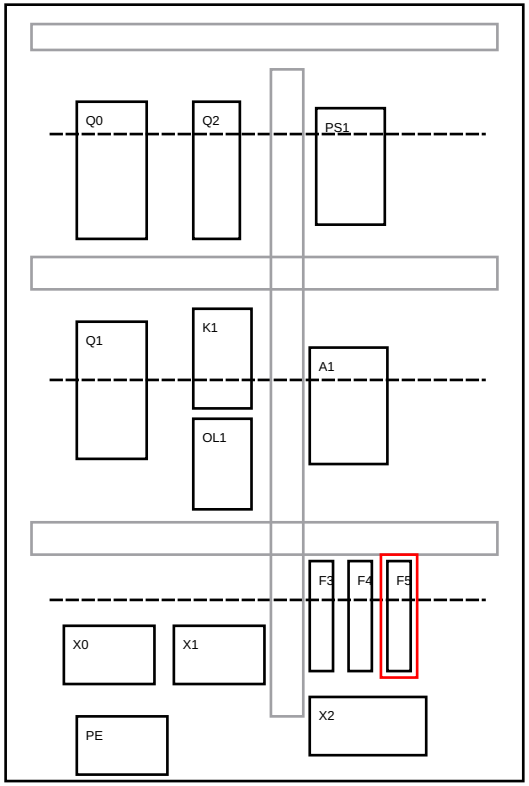
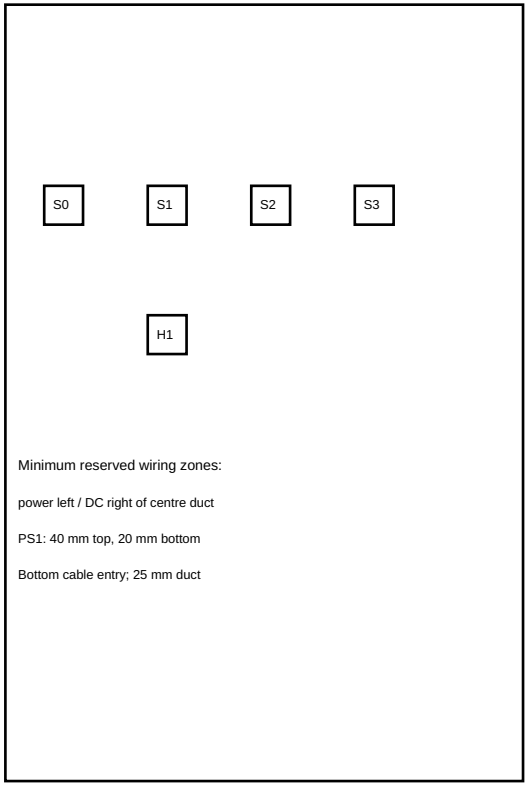
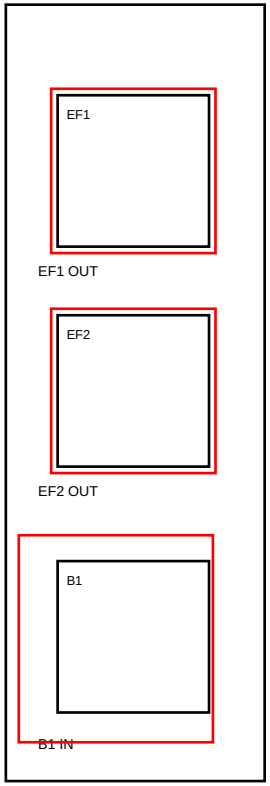


Backplate: 400 W x 600 H; enclosure depth 200



Door: 400 W x 600 H; bond to PE



Side: 200 D x 600 H

B: added fan + outlet + F5

E1: 600 H x 400 W x 200 D steel. Coordinates below are millimetres from backplate upper-left.

Main power left; 24 V control right. Separate duct routes. DIN rails at y=100, 290, 460 mm.

Envelopes include conservative planning sizes; verify final vendor drawings and wiring clearances before build.

PS1: reserve 40 mm above, 20 below per mounting guidance. Keep fan intake and upper outlet clear.

Door: S0, S1, S2, S3 and H1, 30 mm envelopes. Bond door to PE. No mains devices on door.

REV B: added B1 fan, EF1/EF2 outlets and F5 fuse after 40 C ambient check. Redline: drawings/revisions/.

Reference conductors by net name across sheets. Crossings have no junction unless they share a terminal.

Author: Jack Lai	Panel layout, dimensions in mm	File: bottle-filler
Date:		Folio: 5/5